

Student Result Processing System

Module specification

Module name: input.c

Input: It takes Student details(Student ID,Name,Minor marks,Major marks)

Pre-condition: There should be an input file to read this data.

Logic:

Step 1: Start

Step 2: Read the details of students from an input file and store it in structure.

Step 3: Perform validation of the data.

Step 4: Compute the results of students.

Step 5: Stop

Output:Input data is stored in student structure . The data is passed to validation and computation modules.

Module name: validation.c

Input: It takes Student details(Student ID,Name,Minor marks,Major marks)

Pre-condition: The input read from the file have to be parsed properly.

Logic:

Step 1: Start

Step 2: If student id is not an alphanumeric ,then print invalid id and return 0.

Step 3: Check whether the student ID already exists, if duplicate present , then return 0.

Step 4: if minor marks are not within 0 and 40 ,display invalid and return 0.

Step 5: if major marks are not within 0 and 60 ,display invalid and return 0.

Step 6: Store the id in id list.

Step 7: Return 1 if student id, marks and name is valid

Step 8: Stop

Output:Return the value to input module

Module name: computation.c

Input: It takes validated student details.

Pre-condition: The student id, names and marks should be validated.

Logic:

Step 1: Start

Step 2: Initialize grade =0

Step 3: for each subject calculate the total by adding minor and major marks and display it.

Step 4: if total >= 90

 Display grade as 'O'

 grade += 10

Step 4.1: else if total >= 80

 Display grade as 'A+'

 grade += 9

Step 4.2: else if total >= 70

 Display grade as 'A'

 grade += 8

Step 4.3: else if total >= 60

 Display grade as 'B+'

 grade += 7

Step 4.4: else if total >= 50

 Display grade as 'B'

 grade += 6

Step 4.5: else if total >= 40

 Display grade as 'C'

 grade += 5

Step 4.6: else if

 Display grade as 'F'

End if

End for

Step 5: Calculate cgpa and percentage

 cgpa=grade/5

 percentage=cgpa*10

Step 6: calculate highest, lowest percentage, total percentage and student count of every grade.

Step 7 :Stop

Output:Return the value to input module

Module name: output.c

Input: Class average percentage, highest and lowest percentage, number of students in each grade category.

Pre-condition: There should be an output file to display the output.

Logic: Print the class average percentage, highest and lowest percentage, number of students in each grade category.

Output: Display class average percentage, highest and lowest percentage, number of students in each grade category.